

NISHA RAJPUT

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Profile

Software Engineer with expertise in distributed backend systems, high-throughput ETL pipelines, and cloud-native microservices. Proficient in Python, SQL, Java, and Kubernetes, with a consistent record of building scalable production solutions processing 100K+ records. Collaborative communicator who bridges engineering and business teams to deliver data-driven outcomes.

Employment History

George Mason University

Classroom Technology Assistant / AI & Data Engineer

Fairfax, VA

May 2025 – Present

- Spearheaded AI-driven ticket classification workflows leveraging GPT-4 and LangChain, reducing manual triage overhead by 35% across 40+ enterprise classrooms and elevating operational throughput.
- Constructed high-throughput ETL pipelines ingesting and transforming 10,000+ operational log entries monthly, enhancing analytics query readiness and curtailing manual reporting effort by 25%.
- Fortified service reliability by elevating SLA adherence to 35% through comprehensive escalation redesign and systematic incident-resolution protocols, communicating status updates to stakeholders throughout.
- Orchestrated deployment of 5+ containerized backend services on Kubernetes clusters, accelerating release cadence by 30% and bolstering long-term deployment stability.

ExcelR Pvt. Ltd

Full Stack Developer Intern

Hyderabad, India

Aug 2023 – Aug 2024

- Synthesized requirements from 3 stakeholder cohorts into 8+ scalable web modules utilizing ReactJS, Node.js, and SQL, aligning application logic with measurable KPI objectives and presenting progress in structured sprint reviews.
- Overhauled relational schema design and optimized SQL queries across 50,000+ user interaction records, slashing query execution latency by 30% and enabling real-time performance monitoring.
- Supervised validation cycles for 6 platform releases, formulating structured test scenarios that guaranteed 100% alignment between functional outputs and business expectations.
- Pioneered analytics-informed interface enhancements that amplified user engagement by 20% and strengthened session-retention performance across key product surfaces.

Academic Projects

AURA: Multimodal Emotion Recognition System — AI/ML Engineering

- Devised a multimodal classification pipeline leveraging DistilRoBERTa and Wav2Vec2 to process 15,000+ labeled audio-text samples, achieving 88% predictive accuracy.
- Exposed RESTful inference endpoints sustaining 500+ concurrent evaluation requests with sub-second latency, demonstrating robust scalability under load.
- Refined preprocessing and feature-extraction workflows, diminishing model evaluation overhead by 25

Document Workflow Automation & ROI Analysis — Business Case & Strategy Project

- Scrutinized 5-stage operational workflows to diagnose inefficiencies in cross-departmental document routing, presenting findings to project stakeholders via structured executive summaries.
- Restructured process architecture by eliminating 40% redundant handling steps, markedly accelerating task throughput and reducing inter-team friction.
- Projected financial impact estimating 780+ annual hours recovered and a 22% operational cost reduction across cross-functional document workflows.

E-Commerce KPI Dashboard via SQL Extraction— Data Analytics Project

- Aggregated and interrogated 100,000+ transactional records through advanced SQL (joins, aggregations, CTEs), generating actionable revenue intelligence for multi-region performance analysis.
- Employed exploratory analytics to uncover high-value customer segments, enhancing projected targeting precision by 18%.
- Devised optimization strategies reallocating marketing investment across 4 geographic regions, grounding recommendations in data-driven evidence.

Skills

Programming Languages/Web Technologies: HTML, CSS, Python, JavaScript

Databases: MySQL, MongoDB

Cloud & DevOps: Docker, Kubernetes, Jenkins, AWS

Business Analytics Tools: Tableau, Power BI, Jira, Advanced Excel (PivotTables, Power Query), KPI Development, Data Visualization, Requirements Analysis

Version Control: Git

Education

Master of Science (MS) – Computer Science, *George Mason University*

Aug 2024 – May 2026

CGPA: 3.37

Bachelor of Technology (B.Tech) – Computer Science, *Vidya Jyothi Institute of Technology*

May 2021 – Apr 2024

CGPA: 3.41

Certifications

- Qlik Sense Business Analyst Qualification 2023 – Qlik (Issued: September 2023)
- Python Essentials Certification (PCAP – Programming Essentials in Python) – Cisco Networking Academy
- Oracle Database Management System (DBMS) Certification – Oracle Corporation